

Skill-8

To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
srivalli@DESKTOP-6S9T6SD: ~$ cd ~
srivalli@DESKTOP-6S9T6SD: ~$ mkdir OS_Skill8
srivalli@DESKTOP-6S9T6SD: ~$ cd OS_Skill8
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ pwd
/home/srivalli/OS_Skill8
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ ls
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ nano variable_expansion.c
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ gcc -Wall -Wextra variable_expansion.c -o variable_expansion
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ ls
variable_expansion  variable_expansion.c
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ ./variable_expansion
Variable Expansion Test
Enter a string containing variables:
Hello $USER

Original: Hello $USER
Expanded: Hello srivalli
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ ./variable_expansion
Variable Expansion Test
Enter a string containing variables:
Home directory: $HOME

Original: Home directory: $HOME
Expanded: Home directory: /home/srivalli
srivalli@DESKTOP-6S9T6SD: ~/OS_Skill8$ ./variable_expansion
Variable Expansion Test
Enter a string containing variables:
Value: $UNDEFINED_VAR

Original: Value: $UNDEFINED_VAR
Expanded: Value:
```

```
srivalli@DESKTOP-6S9T6SD: ~ x + v
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ ./variable_expansion
Variable Expansion Test
Enter a string containing variables:
User=$USER, Home=${HOME}

Original: User=$USER, Home=${HOME}
Expanded: User=srivalli, Home=/home/srivalli
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ ./variable_expansion
Variable Expansion Test
Enter a string containing variables:
Path=$PATH

Original: Path=$PATH
Expanded: Path=
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ nano builtin_commands.c
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ gcc -Wall -Wextra builtin_commands.c -o builtin_commands
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ ls
builtin_commands  variable_expansion
builtin_commands.c  variable_expansion.c
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ ./builtin_commands
Built-in Command Dispatch Test

mini-shell> pwd
Current directory: /home/srivalli/OS_Skill8

mini-shell> help

Built-in commands:
pwd  - Display current directory
cd   - Change directory
help - Display available commands
exit - Exit the program

mini-shell> cd ..

mini-shell> pwd
Current directory: /home/srivalli
```

```
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ ./builtin_commands  
Built-in Command Dispatch Test
```

```
mini-shell> pwd  
Current directory: /home/srivalli/OS_Skill8
```

```
mini-shell> help
```

```
Built-in commands:  
  pwd  - Display current directory  
  cd   - Change directory  
  help - Display available commands  
  exit - Exit the program
```

```
mini-shell> cd ..
```

```
mini-shell> pwd  
Current directory: /home/srivalli
```

```
mini-shell> hello  
Invalid command: hello
```

```
mini-shell> exit  
Exiting built-in command test.
```

```
srivalli@DESKTOP-6S9T6SD:~/OS_Skill8$ |
```